



SYDNEY BOYS HIGH
MOORE PARK, SURRY HILLS

2003

YEAR 11 YEARLY EXAMINATION

Mathematics

General Instructions

- Reading time — 5 minutes
- Working time — 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- All necessary working should be shown in every question
- Start each Section in a new booklet

Total marks — 130

- Attempt all questions
- Questions are not of equal value, the mark value is shown beside each part.
- Hand up your paper in five parts:
Section A, Questions 1 & 2;
Section B, Questions 3 & 4;
Section C, Questions 5 & 6;
Section D, Questions 7 & 8;
Section E, Questions 9 & 10

Examiner: R.Boros

Total marks – 130
Attempt Questions 1–10

Answer each Section in a SEPARATE writing booklet. Extra writing booklets are available upon request.

Section A Use a SEPARATE writing booklet.

Marks

Question 1 (16 marks)

(a) Express each of the following as a rational number.

(i) $49^{-\frac{1}{2}} \times 27^{\frac{2}{3}}$ 1

(ii) The quotient of $\sqrt{7}$ and $\sqrt{63}$ 1

(iii) $\log_2 8$ 1

(iv) $\frac{\sqrt{32} - \sqrt{8}}{3\sqrt{2}}$ 2

(b) Find, correct to 2 decimal places, $\frac{(3 \cdot 24)^2}{5 \cdot 73 - 2 \cdot 84}$. 1

(c) Solve for x , $\frac{2x}{x-5} = \frac{3}{5}$. 2

(d) Factorise fully the expression $x^3 - x^2 - x + 1$. 2

(e) Find the centre and radius of the circle $x^2 + y^2 - 6x + 4y - 12 = 0$. 2

(f) Solve the inequality $x^2 - 4x < 0$. 2

(g) Given $v^2 = u^2 - 2aS$, $v = 2 \cdot 5$, $u = 2 \cdot 3$, and $a = 7$, find S correct to 3 significant figures. 2

Section A continued

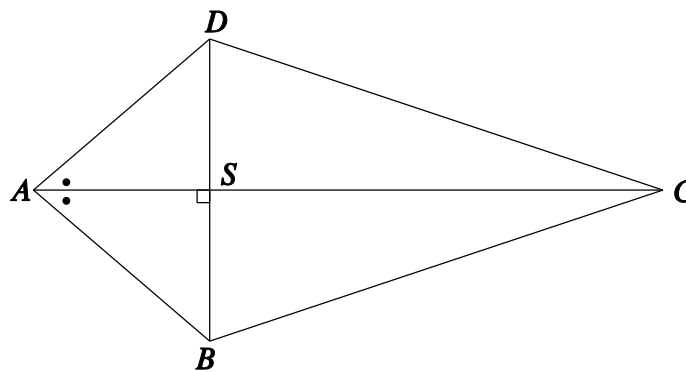
Marks

Question 2 (14 marks)

(a) Given the points $A(-5, 3)$, $B(1, -5)$, and $C(2, 2)$,

- | | |
|---|----------|
| (i) Find the length of AB . | 1 |
| (ii) Find the equation of the line AB written in general form. | 2 |
| (iii) Find the perpendicular distance of C from the line AB . | 2 |
| (iv) Hence or otherwise, calculate the area of the triangle ABC . | 1 |

(b)



In the above diagram, not to scale, $ABCD$ is a quadrilateral. The diagonals AC and DB intersect at right angles at point S . $\hat{DAS} = \hat{BAS}$.

- | | |
|--|----------|
| (i) Prove that $\triangle ASB$ is congruent to $\triangle ASD$. | 2 |
| (ii) Hence prove that $DA = BA$. | 1 |
- (c) Given $f(x) = x^2 + 3x + 2$,
- | | |
|--|----------|
| (i) Evaluate $f(-3)$. | 1 |
| (ii) Find a simple expression for $f(a+2)$. | 2 |
- (d) Solve for x , $|2x - 1| \leq 5$.
- 2**

Section B Use a SEPARATE writing booklet.

Marks

Question 3 (13 marks)

- (a) Doug observes a clifftop A from his yacht at position P . He then sails 500 m towards the cliff to position Q . The angle of elevation to the clifftop from P is 5° and from Q is 8° .
- (i) Draw a diagram to illustrate the above information and use the Sine Rule to calculate AQ correct to the nearest metre. 2
- (ii) Hence or otherwise find the distance QB correct to the nearest 10 metres. 2
- (b) Simplify the expression $\frac{\tan\theta}{\cot\theta} - \frac{\sec^2\theta}{1}$. 2
- (c) (i) Find θ given that $\sin\theta = \frac{1}{2}$ where $0^\circ \leq \theta \leq 180^\circ$. 1
- (ii) Hence find the exact values of $\tan \theta$ and $\sec \theta$. 2
- (d) Simplify $\sin\theta \cos(90^\circ - \theta) + \cos\theta \sin(90^\circ - \theta)$. 2
- (e) If $\sin\theta = \frac{8}{17}$ and θ is an acute angle, find the exact values of $\cos \theta$ and $\tan \theta$. 2

Section B continued**Marks****Question 4** (15 marks)

- (a) Sketch the area defined by the inequality $y \leq x^2$. **2**
- (b) Let α and β be the roots of the equation $x^2 - 5x + 2 = 0$. Find the values of:
- (i) $\alpha + \beta$ **1**
- (ii) $\alpha \beta$ **1**
- (iii) $(\alpha + 1)(\beta + 1)$ **1**
- (c) State the domain and range of each of these functions.
- (i) $x^2 + 3 = y$ **2**
- (ii) $y = 3^x$ **2**
- (d) Solve for x the equation $9^x - 9(3)^x = 0$. **3**
- (e) The roots of the quadratic equation $px^2 - x + q = 0$ are -1 and 3 . Find p and q . **3**

Section C Use a SEPARATE writing booklet.

Marks

Question 5 (15 marks)

- (a) A parabola has the equation $y = x^2 - 12x + 20$. Find
- (i) where it cuts the x and y axes, 2
 - (ii) its axis of symmetry and vertex, 2
 - (iii) the focus, by first expressing it in the form $(x - h)^2 = 4a(y - k)$, 2
 - (iv) the equation of the directrix. 1
- (b) For each of the quadratics below, evaluate the discriminant and state the relevance of this with regard to the roots of the equation.
- (i) $x^2 + 3x + \frac{9}{4} = y$ 2
 - (ii) $3x^2 - 2x = y + 5$ 2
- (c) Find the values of M for which the equation $4x^2 - Mx + 9 = 0$ has
- (i) exactly one real root, 2
 - (ii) real roots. 2

Section C continued**Marks****Question 6** (14 marks)

- (a) Solve the following equations simultaneously,
 $4x - y = 3$ and $10x + 3y = 2$. **2**
- (b) Find x given $2\log_9\sqrt{3} + \log_9 81 = x$. **3**
- (c) Find x correct to 3 decimal places given that $7^x = 15$. **2**
- (d) If Ron invests \$500 at 12.5 % p.a. compound interest, how long would it take the investment to grow to a sum of \$1000. (Answer in years, correct to 2 decimal places.) **3**
- (e) The r^{th} term of a series is $3 \times 2^{(r-4)}$. Determine which of the numbers 96, $\frac{3}{4}$, 256 belong to the series. **2**
- (f) Evaluate $\sum_{n=-1}^7 (2n + 3)$. **2**

Section D Use a SEPARATE writing booklet.

Marks

Question 7 (10 marks)

- (a) Let $A (0, -2)$ and $B (1, 0)$ be 2 fixed points and let $P (x, y)$ be a variable point. **3**
Find the locus of P such that the length $(PA)^2$ equals the length $(PB)^2$.
- (b) Find the locus of points 2 units away from the line $y = 3$. **2**
- (c) Sketch the graph of $y = \sin \theta$ given $-180^\circ < \theta < 90^\circ$. **3**
- (d) Find the values of a and b given that **2**
$$(ax - 3)^2 + b = 4x^2 - 12x + 15.$$

Section D continued**Marks****Question 8** (12 marks)

(a) Differentiate each of the following:

(i) $y = 2x^3 - 8$ **1**

(ii) $y = (2x - 1)^3$ **1**

(iii) $y = \frac{2x}{1-3x}$ **2**

(iv) $y = x^2\sqrt{x}$ **2**

(v) $y = \frac{7}{2x^3}$ **1**

(b) (i) Find the gradient of the normal to the curve $y = 1 - \frac{1}{2}x^2$ at the point (1, 3). **3**

(ii) Find the point on $y = 1 - \frac{1}{2}x^2$ where the tangent to the curve is parallel to this normal. **2**

Section E Use a SEPARATE writing booklet.

Marks

Question 9 (11 marks)

(a) Evaluate the following limits:

(i) $\lim_{x \rightarrow 5} \left(\frac{x-5}{x^2-25} \right)$ **2**

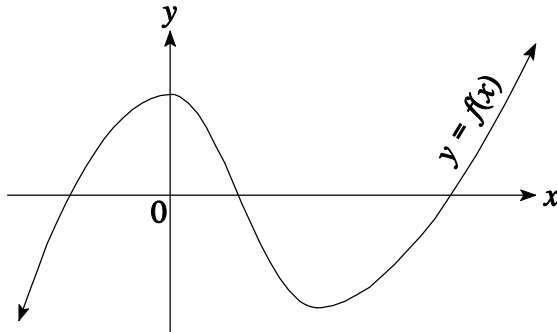
(ii) $\lim_{x \rightarrow \infty} \left(\frac{7-2x-3x^2}{5x^2+3} \right)$ **3**

(b) Two cars depart town *A* at the same time. Car *X* travels at 60 km/h on a bearing of 345°T , whilst car *Y* travels at 100 km/h on a bearing of 085°T . How far apart would the cars be after 3 hours? Answer correct to one decimal place. **3**

(c) Differentiate $y = x^2 + x$ from first principles. **3**

Section E continued**Marks****Question 10** (10 marks)

- (a) The diagram below shows $y = f(x)$.

3

Copy or trace this diagram into your answer booklet and sketch a possible graph for $y = f'(x)$ on the same set of axes.

- (b) A farmer wishes to make a rectangular enclosure using a river as one boundary and 400 m of fencing on the other three sides.

- (i) Find the maximum possible area of the enclosure.

3

- (ii) What are the dimensions of this enclosure?

1

- (c) Find x correct to 3 decimal places, given that $\log_7 6 - 2\log_7 3 = x$.

3**END OF THE PAPER**



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

SEPTEMBER 2003

YEARLY EXAMINATION

YEAR 11

Mathematics

Sample Solutions

Question (1) [16 marks]

(a) $49^{-1/2} \times 27^{2/3}$

(i) $= \frac{1}{\sqrt{49}} \times (3^3)^{2/3}$ ①

(ii) $\frac{\sqrt{7}}{\sqrt{63}} = \frac{1}{3}$ ①

(iii) $\log_2 8 = 3 \log_2 2 = 3$ ①

(iv) $\frac{2\sqrt{2}(2-1)}{3\sqrt{2}} = \frac{2}{3}$ ①

(b) $\frac{(3 \cdot 24)^2}{5 \cdot 73 - 2 \cdot 84} = \frac{3 \cdot 63}{(2d.f.)}$ ①

(c) $\frac{2x}{x-5} = \frac{3}{5}$

$10x = 3x - 15$

$7x = -15$ ②

(d) $\frac{x^2(x-1) - (x-1)}{(x-1)^2(x+1)}$ ②

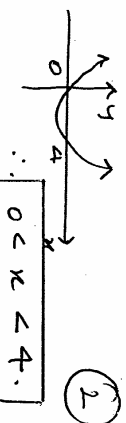
(e) $x^2 - 6x + 9 + y^2 + 4y + 4 = 25$ ②

$\therefore (x-3)^2 + (y+2)^2 = 25$

Centre (3, -2) r = 5

(f) $x^2 - 4x < 0$

$x(x-4) < 0$



(g) $(2.5)^2 = (2.3)^2 - 14(s)$ ②

$14s = 0.96$

Question (2) [14 marks]

(a) (i) $AB = \sqrt{(-6)^2 + 8^2} = 10$ ①

(ii) $w = -\frac{8}{6} = -\frac{4}{3}$

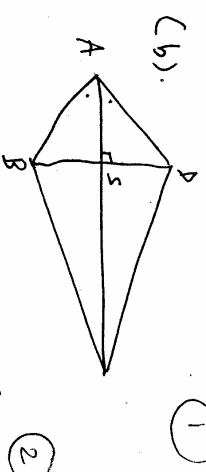
$y - 3 = -\frac{4}{3}(x + 5)$ ②

$3y - 9 = -4x - 20$

$4x + 3y + 11 = 0$

(iii) $d = \frac{1}{\sqrt{25}} \sqrt{8^2 + 11^2} = \frac{\sqrt{173}}{5}$ ②

(iv) $A = \frac{1}{2} \times 10 \times \frac{2\sqrt{5}}{5} = 2\sqrt{5}$ ①



$\angle DAS = \angle BAS$ (given) ②

$\angle DSA = \angle BSA$ (st. line)

AS is common

$\therefore \triangle ASB \cong \triangle ASD$ (AAS)

$DA = BA$

(every sides of $\triangle ASD$)

(c) $f(x) = x^2 + 3x + 2$

(i) $f(-3) = 2$ ①

(ii) $(a+2)^2 + 3(a+2) + 2 = a^2 + 7a + 12$ ②

(d) $(a+3)(a+d)$

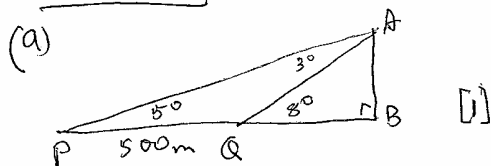
$2x - 1 \leq 5$

$-5 \leq 2x - 1 \leq 5$

$-4 \leq 2x \leq 6$

$-2 \leq x \leq 3$ ②

Question 3



(i) $\angle PAQ = 3^\circ$ (ext. angle thm)

$$\frac{AQ}{\sin 5^\circ} = \frac{500}{\sin 3^\circ}$$

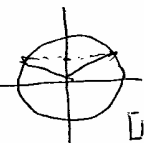
$$AQ = \frac{500 \sin 5^\circ}{\sin 3^\circ} \approx 833 \text{ m} \quad [1]$$

(ii) $\cos 8^\circ = \frac{QB}{AQ}$

$$\therefore QB = AQ \cos 8^\circ \approx 820 \text{ m (nearest 10m)} \quad [2]$$

(b) $\frac{\tan \theta}{\cot \theta} - \sec^2 \theta$
 $= \tan^2 \theta - \sec^2 \theta$
 $= \tan^2 \theta - (1 + \tan^2 \theta)$
 $= -1 \quad [2]$

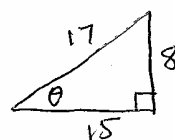
(c) (i) $\sin \theta = \frac{1}{2} \quad 0^\circ \leq \theta \leq 180^\circ$

$\theta = 30^\circ \text{ or } 150^\circ$ 

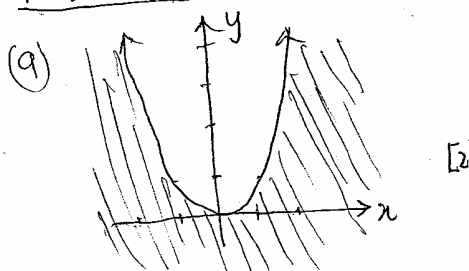
(ii) $\tan \theta = \frac{1}{\sqrt{3}} \text{ or } -\frac{1}{\sqrt{3}}$
 $\sec \theta = \frac{2}{\sqrt{3}} \text{ or } -\frac{2}{\sqrt{3}} \quad [2]$

(d) $\sin \theta \cdot \cos(90^\circ - \theta) + \cos \theta \sin(90^\circ - \theta)$
 $= \sin \theta \cdot \sin \theta + \cos \theta \cdot \cos \theta$
 $= \sin^2 \theta + \cos^2 \theta \quad [2]$
 $= 1$

(e) $\sin \theta = \frac{8}{17}$
 $\cos \theta = \frac{15}{17} \quad [2]$
 $\tan \theta = \frac{8}{15}$



Question 4



(b) $x^2 - 5x + 2 = 0$

(i) $\alpha + \beta = 5 \quad [1]$

(ii) $\alpha \beta = 2 \quad [1]$

(iii) $(\alpha + 1)(\beta + 1) = \alpha\beta + (\alpha + \beta) + 1$
 $= 2 + 5 + 1$
 $= 8 \quad [1]$

(c) (i) $y = x^2 + 3 \quad D = \{x: x \in \mathbb{R}\}$
 $R = \{y: y \geq 3\}$

(ii) $y = 3^x \quad D = \{x: x \in \mathbb{R}\}$
 $R = \{y: y > 0\}$

Question 4 (cont'd)

(d) $9^x - 9(3)^x = 0$

$(3^x)^2 - 9(3^x) = 0$

Let $u = 3^x$

$u^2 - 9u = 0$

$u(u - 9) = 0$

$\therefore u = 0$ or $u = 9$

$\therefore 3^x = 0$ or $3^x = 9$

No soln $\underline{\underline{x = 2}}$ [3]

(e) $px^2 - x + q = 0$

Roots -1 and 3

$\alpha + \beta = \frac{1}{p}$ $\alpha \beta = \frac{q}{p}$

$\therefore \frac{1}{p} = 2$	$\frac{q}{p} = -3$
$p = \frac{1}{2}$	$2q = -3$
	$q = -\frac{3}{2}$

[3]

- [2] 5. (a) (i) On the x -axis, $y = 0$; so $(x - 10)(x - 2) = 0$.
On the y -axis, $x = 0$; so $y = 20$.
 $\therefore$ The curve meets the x -axis at 2, 10, and the y -axis at 20.
- [2] (ii) The midpoint of 2 and 10 is 6, so the axis of symmetry is $x = 6$.
When $x = 6$, $y = -16$, so the vertex is $(6, -16)$.
- [2] (iii) $x^2 - 12x + 36 = y - 20 + 36$,
 $(x - 6)^2 = 4 \times \frac{1}{4} \times (y + 16)$.
 $\therefore$ The focus is at $(6, -15\frac{3}{4})$.
- [1] (iv) The directrix is $y = -16\frac{1}{4}$.
- [2] (b) (i) $\Delta = 9 - 9 = 0 \implies$ one real root.
- [2] (ii) $\Delta = 4 + 60 = 64 \implies$ two real, rational roots.
- [2] (c) (i) $\Delta = M^2 - 144 = 0$,
 $M^2 = 144$.
 $\therefore$ One root when $M = \pm 12$.
- [2] (ii) $M^2 - 144 \geq 0$,
 $M^2 \geq 144$,
 $|M| \geq 12$.
 $\therefore M \leq -12, M \geq 12$.
- [2] 6. (a) $4x - y = 3, \dots\dots\dots(1)$
 $10x + 3y = 2. \dots\dots\dots(2)$
 $y = 4x - 3$ from (1), then substitute in (2).
 $10x + 3(4x - 3) = 2$,
 $10x + 12x - 9 = 2$,
 $22x = 11$,
 $x = \frac{1}{2}$,
 $y = -1$.
- [3] (b) $\log_9 3 + \log_9 9^2 = x$,
 $\log_9 9^{\frac{1}{2}} + \log_9 9^2 = x$,
 $\frac{1}{2} \log_9 9 + 2 \log_9 9 = x$,
 $\frac{1}{2} + 2 = x$,
i.e., $x = 2\frac{1}{2}$ (or $\frac{5}{2}$).
- [2] (c) $\log 7^x = \log 15$,
 $x \log 7 = \log 15$,
 $x = \frac{\log 15}{\log 7}$,
 ≈ 1.392 .

$$\begin{aligned}
 \boxed{3} \quad (d) \quad & \$1\,000 = \$500 \left(1 + \frac{12.5}{100}\right)^n, \\
 & 2 = (1.125)^n, \\
 & \log 1.125^n = \log 2, \\
 & n \log 1.125 = \log 2, \\
 & n = \frac{\log 2}{\log 1.125} \\
 & n \approx 5.88 \text{ years.}
 \end{aligned}$$

$$\begin{aligned}
 \boxed{2} \quad (e) \quad & 96 \div 3 = 32, & \frac{3}{4} \div 3 = \frac{1}{4}, & 256 \div 3 = 85\frac{1}{3}, \\
 & = 2^5. & = 2^{-2}. & \text{Not a power of 2.} \\
 & \therefore 96, \frac{3}{4} \text{ belong to the series.}
 \end{aligned}$$

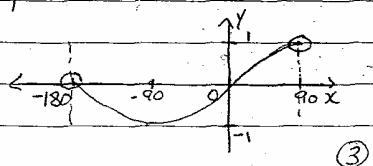
$$\begin{aligned}
 \boxed{2} \quad (f) \quad & \sum_{n=-1}^7 (2n+3) = 1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17, \\
 & = 81.
 \end{aligned}$$

QUESTION 7

(a) $A(0, -2) B(1, 0) (PA)^2 = (PB)^2$
 $(x-0)^2 + (y+2)^2 = (x+1)^2 + (y-0)^2$
 $x^2 + y^2 + 4y + 4 = x^2 + 2x + 1 + y^2$
 $4y + 2x + 3 = 0$
 $2x + 4y + 3 = 0$ (3)

(b) $y = 5$ or $y = 1$ (2)

(c) $y = \sin \theta \quad -180^\circ < \theta < 90^\circ$



(d) $(ax-3)^2 + b = 4x^2 - 12x + 15$
 $a^2x^2 - 6ax + 9 + b = 4x^2 - 12x + 15$
 equating coefficients
 $a^2 = 4 \quad -6ax = -12x \quad 9+b = 15$
 $a = \pm 2 \quad a = 2 \quad b = 6$
 $a = 2, b = 6$
 or
 $(ax-3)^2 + b = 4x^2 - 12x + 9 + 6$
 $(ax-3)^2 + b = (2x-3)^2 + 6$
 $a = 2 \quad b = 6$ (2)

QUESTION 8

(a)(i) $y = 2x^3 - 8$
 $y' = 6x^2$ (1)

(ii) $y = (2x-3)^3$
 $y' = 3(2x-3)^2 \times 2$
 $= 6(2x-3)^2$ (1)

(iii) $y = \frac{2x}{1-3x}$
 $y' = \frac{vdu - u dv}{v^2}$

$= \frac{(1-3x) \times 2 - 2x \times -3}{(1-3x)^2}$

$= \frac{2}{(1-3x)^2}$ (2)

(iv) $y = x^2 \sqrt{x}$
 $= x^{\frac{5}{2}}$
 $y' = \frac{5}{2} x^{\frac{3}{2}}$
 $= \frac{5}{2} \sqrt{x^3}$ (2)

(v) $y = \frac{7}{2x^3} = \frac{7}{2} x^{-3}$
 $y' = -\frac{21}{2} x^{-4}$ (1)
 $= -\frac{21}{2x^4}$

(b) $y = 1 - \frac{1}{2} x^2$
 (i) $y' = -x$
 when $x = 1$ grad of tangent = -1
 " normal = 1 (3)

(ii) $y' = -x$
 $-x = 1$ when $x = -1$
 point $(-1, \frac{1}{2})$ (2)

Question 9.

$$\begin{aligned}
 \text{(a) (i)} \quad & \lim_{x \rightarrow 5} \frac{x-5}{x^2-25} \\
 &= \lim_{x \rightarrow 5} \frac{x-5}{(x-5)(x+5)} \\
 &= \lim_{x \rightarrow 5} \frac{1}{(x+5)} \\
 &= \frac{1}{10}
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad & \lim_{x \rightarrow \infty} \frac{7-2x-3x^2}{5x^2+3} \\
 &= \lim_{x \rightarrow \infty} \frac{\frac{7}{x^2} - \frac{2}{x} - \frac{3x^2}{x^2}}{\frac{5x^2}{x^2} + \frac{3}{x^2}} \\
 &= -\frac{3}{5}
 \end{aligned}$$

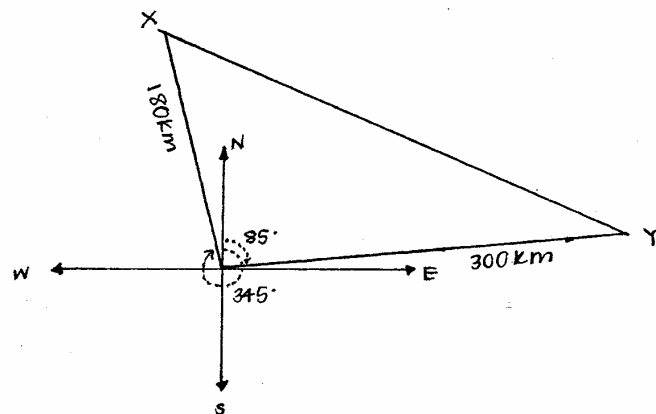
- (b) In 3 hours car X travels 180km.
In 3 hours car Y travels 300km.

$$\begin{aligned}
 XY^2 &= AY^2 + AX^2 - 2AY \cdot AX \cdot \cos A \\
 &= 300^2 + 180^2 - 2(300)(180)\cos 100^\circ
 \end{aligned}$$

$$\begin{aligned}
 XY^2 &= AY^2 + AX^2 - 2AY \cdot AX \cdot \cos A \\
 &= 300^2 + 180^2 - 2(300)(180)\cos 100^\circ
 \end{aligned}$$

$$XY = 375.7$$

$$\therefore XY = 375.7 \text{ km (1d.p.)}$$



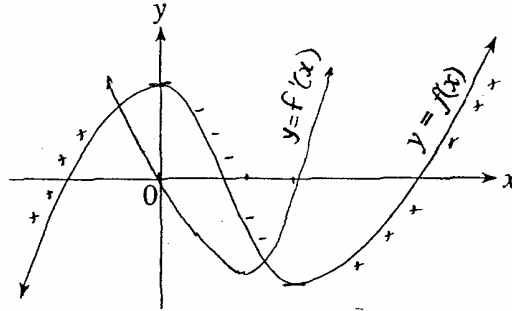
(c) Let $f(x) = x^2 + x$
 Let $f(x+h) = (x+h)^2 + (x+h)$

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow \infty} \frac{f(x+h) - f(x)}{h} \\
 &= \lim_{h \rightarrow \infty} \frac{(x+h)^2 + (x+h) - (x^2 + x)}{h} \\
 &= \lim_{h \rightarrow \infty} \frac{x^2 + 2xh + h^2 + x + h - x^2 - x}{h} \\
 &= \lim_{h \rightarrow \infty} \frac{2xh + h^2 + h}{h} \\
 &= \lim_{h \rightarrow \infty} \frac{h(2x + h + 1)}{h} \\
 &= 2x + 1
 \end{aligned}$$

$\therefore f'(x) = 2x + 1$ by first principles.

Question 10.

(a)



(b) (i) Let $P = 2x + y$, where $P = 400$.

$$400 = 2x + y$$

$$\therefore y = 400 - 2x$$

$$\text{Area} = xy$$

$$= x(400 - 2x)$$

$$= 400x - 2x^2$$

Using the axis of symmetry formula, $x = -\frac{b}{2a}$, where $a = -2, b = 400$.

$$\begin{aligned} x &= -\frac{b}{2a} \\ &= -\frac{400}{2(-2)} \\ &= 100 \end{aligned}$$

At $x = 100, y = 200$.

Hence the maximum possible area of this enclosure is $A = 20000 \text{ m}^2$

(ii) Hence the dimensions of this enclosure are 100m by 200m.

(c)

$$\log_7 6 - 2\log_7 3 = x$$

$$\log_7 6 - \log_7 3^2 = x$$

$$\frac{\log_{10} 6}{\log_{10} 7} - \frac{\log_{10} 9}{\log_{10} 7} = x$$

$$0.9207 - 1.1291 = x$$

$$x = -0.20845$$

$$\therefore x = -0.208 \text{ (3dp).}$$

or

$$\log_7 \left(\frac{6}{3^2} \right) = x$$

$$\log_7 \left(\frac{2}{3} \right) = x$$

$$7^x = \frac{2}{3}$$

$$x \log 7 = \log \frac{2}{3}$$

$$\therefore x = -0.208$$